



Center for Geographic Analysis
Institute for Quantitative Social Science
Harvard University

Title: Getting Started with AI for Spatial Data

Instructors: Jielu Zhang and Devika Jain

Time: 9:00 AM–12:00 PM, May 22, 2026

Location: 1737 Cambridge St., Cambridge, MA, CGIS Knafel Building, Room SB-12, Sub-Basement

Zoom: <https://harvard.zoom.us/j/2304706732?omn=92517624341>

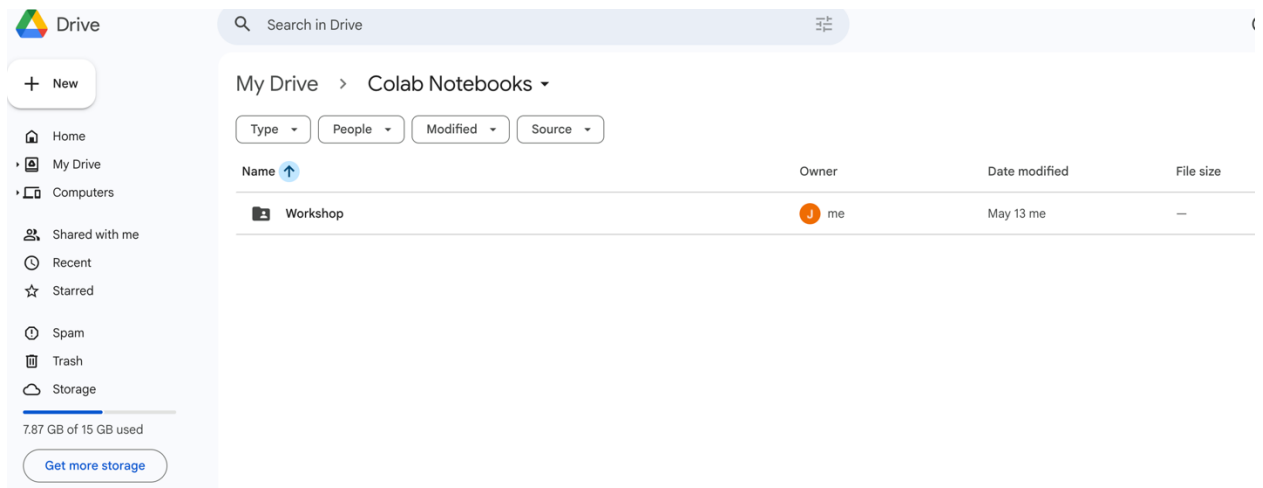
Pre-Workshop Instructions

Laptops are required for this workshop. Please complete the following steps before the workshop begins:

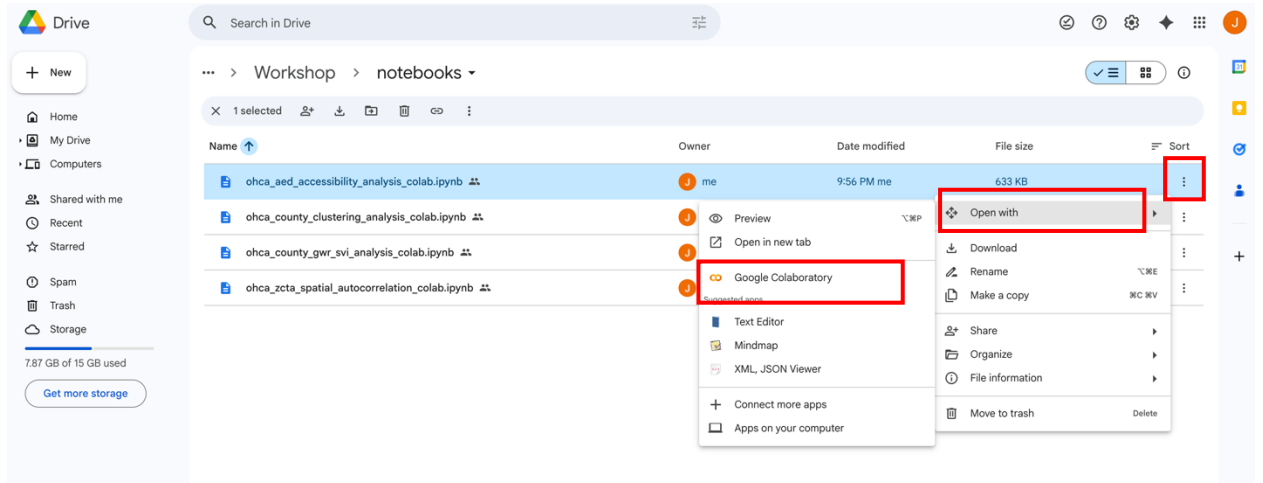
1. Download the GitHub repository for this workshop to your local computer.
2. Log in to Google Colab using your Gmail account:
<https://colab.research.google.com/>
3. Mount your Google Drive in Google Colab.

The screenshot shows the Google Colab web interface. At the top, it says 'Welcome To Colab' and 'Cannot save changes'. Below the menu bar (File, Edit, View, Insert, Runtime, Tools, Help), there are tabs for 'Commands', '+ Code', '+ Text', and a 'Run all' button. On the left, a 'Files' panel is open, showing a file explorer view. A red box highlights the 'upload' icon (a folder with an arrow) and is labeled 'Step 1'. Another red box highlights the 'drive' icon (a folder with a car) and is labeled 'Step 2'. In the main code area, a code cell is shown with the text: `from google.colab import drive` and `drive.mount('/content/drive')`. A blue tooltip box is overlaid on the code cell, saying 'Run this cell to mount your Google Drive.' with a 'Dismiss' button. Below the code cell, there is a message: 'Google Colab is available in VS Code!' with logos for Colab, a heart, and VS Code. At the bottom, there is a text block: 'Try the new [Google Colab extension](#) for Visual Studio Code. You can get up and running' followed by two bullet points: '• In VS Code, open the **Extensions** view and search for 'Google Colab' to install.' and '• Open the kernel selector by creating or opening any `.ipynb` notebook file in yo clicking the **Select Kernel** button in the top right

4. Open your Google Drive and upload the workshop folder to:
My Drive / Colab Notebooks



5. Open the notebook file (.ipynb) using Google Colaboratory.



6. Randomly select one notebook and make sure you can run it successfully. Use Shift + Enter on Windows or Shift + Return on macOS to run each cell. If you are unable to complete this step successfully, no worries—we can work through it together during the

workshop.

ohca_zcta_spatial_autocorrelation_colab.ipynb

File Edit View Insert Runtime Tools Help

Commands + Code + Text Run all

Connect

OHCA Events per County: Spatial Autocorrelation in Google Colab

This notebook uses the OHCA training dataset and U.S. county boundaries to:

1. Convert OHCA records with latitude/longitude into spatial points.
2. Spatially join points to county polygons.
3. Count OHCA events per county.
4. Run Global Moran's I and Local Moran's I / LISA.
5. Map event counts and statistically significant clusters.

Note: Moran's I requires a variable, not just coordinates. Here the variable is `ohca_event_count`, the number of OHCA events in each county.

1. Install Packages

Run this cell in Google Colab. It installs the geospatial and spatial-statistics libraries used in the workshop.

```
!pip -q install geopandas libpysal esda mapclassify matplotlib folium
```

```
from google.colab import drive
drive.mount('/content/drive/')

Mounted at /content/drive/
```